

Why Does Systems Biology Work? The Global Geometry of Wild-Type Parameter Space in Gene Regulatory Networks

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Ordinary differential equation models of gene regulatory networks routinely contain dozens to hundreds of biochemical parameters, yet experimentalists regularly identify parameter sets that reproduce, and even predict, biological behaviour. A naive probability estimate makes this look astronomically unlikely: for one hundred parameters each calibrated to within half of its biologically plausible range, a random draw succeeds with probability about 10^{-30} . Two partial explanations have emerged from previous work. *Sloppiness* tells us that, locally, iso-fit surfaces are highly elongated along combinations of parameters that barely affect model output. *Designability* tells us that, globally, phenotype frequencies in the parameter-to-behaviour map are strongly biased, and wild-type behaviour tends to occupy among the larger neutral sets. Neither explanation closes the gap: sloppiness is local and does not describe how acceptable regions are connected, and designability describes the size of the neutral set but not its arrangement. Here we examine the *global geometry* of the wild-type neutral set in parameter space across twelve benchmark models from Gutenkunst et al. (1). We ask whether this region is localized (a compact pocket) or extended (a structure that reaches through parameter space). We combine three probes: traversal of sloppy Hessian eigendirections, the Nested Neutral Set Estimator (NNSE) of Jörg et al. (2) adapted to continuous parameter spaces, and persistent-homology analysis of the resulting point clouds. The results are consistent across models with an extended, space-spanning geometry rather than a compact one. Together with sloppiness and designability, this offers a three-part account of why large systems-biology models can be fit, with biological implications for robustness, neutral drift, and the design of synthetic biological circuits.

genotype-phenotype maps | systems biology | neutral sets | gene regulatory networks | sloppiness | topological data analysis

Introduction

The budding yeast cell-division cycle model of Chen and colleagues (3) is a system of roughly sixty coupled ordinary differential equations with 156 biochemical parameters, describing everything from gene expression to mitotic spindle formation. Once calibrated, it reproduces a catalogue of experimental observations and predicts the viability and phenotype of more than one hundred genetic mutants that were never used in its construction (3–6). It is one of many success stories of ordinary differential equation (ODE) modelling in systems biology (7–17), and on its face it is puzzling.

The parameter space of the Chen model is vast. Allowing only eight discrete values per parameter, following the robustness-sampling scheme of Chen et al. (3), gives $8^{156} \approx 10^{141}$ combinations, a *hyperastronomical* number in the sense of (18, 19). Even a stripped-down version of the

problem sharpens the puzzle: suppose one must fix $N = 100$ parameters, each to within half of its biologically plausible range. A uniform draw lands in the correct half-interval with probability $1/2$ per parameter, so the probability that all one hundred land simultaneously is

$$p_{\text{naive}} = \left(\frac{1}{2}\right)^{100} \approx 10^{-30}. \quad [1]$$

At one draw per nanosecond over the age of the universe (about 4×10^{17} s, or $\sim 4 \times 10^{26}$ trials), one expects far fewer than the $\sim 10^{30}$ trials required for a single success. Yet Chen-style models are calibrated regularly and generalize to unseen perturbations. This is the puzzle we take up. It has the same structure as Levinthal's paradox for protein folding (20) and the Hoyle-Salisbury paradox for protein evolution (21, 22): a naive combinatorial estimate predicts impossibility, but the landscape has enough structure that the search succeeds.

Sloppiness as a local explanation. The theory of sloppiness (1, 23–25) provides the first partial answer. A model is sloppy when the Hessian of the prediction error at a good fit has a broad eigenvalue spectrum, with a few stiff directions along which the fit degrades sharply and many sloppy directions along which it barely changes. Geometrically, the iso-fit surface around a fitted point is a highly elongated ellipsoid, and its

Significance Statement

Mathematical models of gene regulatory networks often contain more than one hundred biochemical parameters, which makes it puzzling that experimentalists can calibrate them at all. Two partial explanations are available in the literature: sloppiness, which says that acceptable parameter sets form locally elongated regions, and designability, which says that simple biological behaviours occupy an inflated share of parameter space. Neither is sufficient on its own. Here we ask a complementary question about the global geometry of the parameter space that produces wild-type behaviour, and find evidence across twelve benchmark ordinary differential equation models that this region is not a localized blob but an extended structure reaching through the space. This perspective offers a geometric complement to sloppiness and designability and helps explain why fitting and generalization succeed in practice.

A.A.L. and G.B. conceived and designed the study. G.B. implemented all models, algorithms and analyses, and wrote the first draft. C.Q.C. contributed model infrastructure and prior analysis. All authors edited the manuscript and gave final approval for publication.

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effective volume is much larger than a naive isotropic estimate would give. Sloppiness has been shown to be a near-universal feature of systems-biology models (1). It is, however, a local property: it describes the shape of the acceptable region around a specific, already-identified parameter set and for a specific set of measurements (26). It says nothing about how many acceptable regions exist in the full space, or whether they connect.

Designability as a global size explanation. A second partial answer comes from treating the parameter-to-behaviour map as a genotype-phenotype (GP) map (27, 28). In this framework the set of parameter vectors producing the same coarse-grained behaviour is called the *neutral set* of that behaviour, and GP-map work on RNA folding (29–32), protein quaternary structure (33), and Boolean gene regulatory networks (34) has consistently found that simpler phenotypes, as measured by descriptive complexity, occupy disproportionately large shares of genotype space. This is known as *simplicity bias* (27), and in our companion work we have shown it also applies to ODE models of gene regulatory networks (35). Wild-type behaviours in those models lie among the more designable phenotypes, with neutral sets that are orders of magnitude larger than those of typical alternatives. This narrows the gap: the wild type is not a randomly chosen target, and the correct denominator in the naive estimate is not the whole parameter space but the effective fraction that produces simple behaviours. Yet the wild type, as we document below, typically occupies on the order of the top one percent of neutral sets by size rather than dominating the distribution, and so designability alone leaves a quantitative residue: the probability of landing in the wild-type neutral set by uniform sampling is still vanishingly small.

The geometry gap. Sloppiness and designability address two different questions. Sloppiness asks what the neutral set looks like locally; designability asks how large it is globally. A third question is left open, which is the subject of this paper.

Given that the wild-type neutral set is larger than naive but still small compared to the full parameter space, how is it arranged?

The question matters because two neutral sets of the same volume can be geometrically very different (Fig. 1). One extreme is a compact region, well-separated from most of parameter space; an arbitrary starting point for gradient-based fitting may be far from it, and reaching it from a random initialization is not guaranteed. The opposite extreme is an extended structure that reaches through the space, a neutral manifold analogous to the neutral networks of RNA sequence space (28–30, 36). A random starting point in that geometry is generically close to some part of the manifold, and gradient flow along sloppy directions reaches it quickly. The two geometries have the same *size*, but they produce radically different *accessibility*.

The distinction has biological consequences. For *calibration*, an extended neutral manifold explains why fitting succeeds from generic starts and why parameter sets obtained this way generalize: the manifold encodes the structural logic of the circuit rather than a specific combination of parameter values. For *evolution*, extendedness permits neutral drift of biochemical parameters under stabilizing selection (28, 31), with robustness to parametric variation as a straightforward

[FIGURE PLACEHOLDER — Fig. 1: Cartoon of localized vs extended neutral-set geometries in a 2D projection of parameter space. Left: a compact region, with a random starting point (red cross) far from it; gradient descent from the cross may never reach the region. Right: an extended structure of the same total area, with random starting points generically close to some part of it; sloppy-direction flow (arrows) reaches a valid parameter set quickly. The two geometries differ in arrangement, not in size.]

Fig. 1. Two candidate geometries for the wild-type neutral set. (A) A localized, compact region. A random initial parameter vector (red cross) is typically far from the region, and gradient-based fitting from the cross has no guarantee of reaching it. **(B)** An extended structure of the same total volume. Random initial points are generically close to some part of the structure, and motion along sloppy directions (arrows) reaches it quickly. The two panels have the same neutral-set size and very different accessibility.

consequence. For *synthetic biology*, the shape of the manifold suggests which parameter combinations can be varied with minimal phenotypic cost.

This paper. We ask how the wild-type neutral set is arranged in parameter space across twelve benchmark ODE models compiled by Gutenkunst et al. (1). We combine three probes, each addressing a different aspect of the question. First, we traverse the Hessian eigendirections from the wild-type anchor and record the distance along each direction over which the phenotype is preserved. Second, we adapt the Nested Neutral Set Estimator (NNSE) of Jörg et al. (2), originally developed for RNA sequence space, to continuous parameter spaces so that we can sample from the neutral set far from the anchor while following its structure. Third, we apply persistent homology (37–39) to the resulting point clouds to quantify connectedness, loopiness, and extent. Across all twelve models the three probes return a consistent picture in which the wild-type neutral set is extended along many sloppy directions and exhibits topological structure inconsistent with a localized compact region. We discuss this geometry as a third component of a layered answer to the calibration paradox, alongside sloppiness and designability.

Results

The parameter-to-phenotype map. Each of the twelve models (Table ??) defines an ODE system

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}, \boldsymbol{\theta}), \quad [2]$$

where $\mathbf{y}(t) \in \mathbb{R}^m$ is a vector of molecular concentrations and $\boldsymbol{\theta} \in \mathbb{R}^N$ is a vector of N biochemical parameters. We define genotype space $\mathcal{G} \subset \mathbb{R}^N$ as the set of parameter vectors in a biologically reasonable neighbourhood of the wild-type values, following the robustness-sampling convention of Chen et al. (3): each parameter θ_i is allowed to take values $\theta_i^* \cdot f$ with $f \in \{0.25, 0.50, \dots, 1.75, 2.00\}$ in the discrete version, and values in the corresponding continuous interval $[0.25\theta_i^*, 2.00\theta_i^*]$ in the continuous version used for NNSE. For the Chen model this gives $8^{156} \approx 10^{141}$ discrete parameter combinations.

A phenotype is defined by coarse-graining the time series produced by the ODE solver. Following Fink et al. (40),

the *oscillation phenotype* of a parameter vector is the binary string obtained by recording the sign of the derivative of the biologically most relevant output molecule (Table ??) at each of $T/\delta t$ evenly spaced time points. This coarse-graining captures the qualitative oscillatory behaviour of the network and has the useful property that its descriptive complexity can be estimated directly with standard string-compression algorithms (41).

The neutral set of a phenotype $\mathbf{x}$ is the set of all parameter vectors mapping to that phenotype,

$$\mathcal{N}(\mathbf{x}) = \{\theta \in \mathcal{G} : \Phi(\theta) = \mathbf{x}\}, \quad [3]$$

where $\Phi : \mathcal{G} \rightarrow \mathcal{P}$ is the genotype-phenotype map. The central object of this paper is not the size of $\mathcal{N}(\mathbf{x}_{\text{wt}})$ but its *shape*: whether it is localized or extended, whether it has topological structure, and how close random points in $\mathcal{G}$ lie to it.

[Table 1: list of 12 ODE models with key output molecule, number of parameters N , number of variables m . Reuse directly from (35).]

Wild-type phenotypes occupy large but not dominant neutral sets. Before asking about the geometry of $\mathcal{N}(\mathbf{x}_{\text{wt}})$, we establish how large it is relative to other neutral sets in the same model. For each of the twelve models, we sampled 5×10^6 parameter vectors uniformly from the Chen-style eight-point grid and recorded the oscillation phenotype of each. The resulting frequency distributions are strongly skewed (Fig. 2A): a small number of phenotypes account for most of the sampled parameter space, and many phenotypes appear only a handful of times. The distributions are consistent with simplicity bias (27, 31, 32): plotting phenotype frequency against an estimator of Lempel-Ziv complexity (41) produces an approximately triangular envelope (Fig. 2B), with low-complexity phenotypes attaining high frequencies and high-complexity phenotypes confined to the low-frequency tail.

The wild-type phenotype is over-represented in this distribution, but not overwhelmingly so. Across the twelve models, the wild-type neutral set ranks within roughly the top one percent of phenotypes by size, with estimated frequencies between 10^{-3} and 10^{-1} (Fig. 2C). This is orders of magnitude above what the naive estimate in Eq. (1) would give for a uniformly chosen target, but still far from the unit frequency one would need for uniform sampling to be a plausible fitting strategy.

[Results paragraph to be refined once revised rank numbers are locked. Expected quantitative message: wild-type frequencies orders of magnitude above naive but still $< 10^{-1}$, confirming designability is partial.]

These observations set up the rest of the paper. If designability alone accounted for why fitting works, the wild type would need to dominate the frequency distribution by many more orders of magnitude than we observe. It does not, and so the explanation must include something beyond neutral-set size.

Neutral sets extend along sloppy directions. We now probe the shape of $\mathcal{N}(\mathbf{x}_{\text{wt}})$ directly. At the wild-type parameter vector θ^* , we compute the Hessian $H(\theta^*)$ of the prediction-error objective with respect to log-parameters, and diagonalize it to obtain eigenvalues $\lambda_1 \leq \dots \leq \lambda_N$ with eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_N$. The eigenvalue spectrum is broad across all twelve

[FIGURE PLACEHOLDER — Fig. 2: Simplicity bias in the parameter-to-phenotype map. (A) Phenotype-frequency distributions for all 12 models, showing strong skew. (B) Frequency-complexity envelopes for all 12 models, with wild-type marker on each. (C) Wild-type rank summary across models: box plot or bar plot showing that wild-type phenotypes consistently fall in the top $\sim 1\%$ by neutral-set size but not at rank 1.]

Fig. 2. Wild-type phenotypes occupy large but not dominant neutral sets. (A) Phenotype frequency distributions under uniform sampling of parameter space for all twelve models. Frequencies span several orders of magnitude, as expected from simplicity bias. (B) Frequency versus Lempel-Ziv complexity for the oscillation phenotype in each model. The approximately triangular upper envelope is a signature of simplicity bias. Wild-type phenotypes (coloured markers) consistently lie close to the upper edge at low complexity. (C) Wild-type rank among all observed phenotypes, by neutral-set size, across the twelve models. The wild type lies in the top percentile of neutral-set sizes but is not the single largest. Designability therefore narrows, but does not close, the gap to the naive estimate.

models, spanning six to eight orders of magnitude (Fig. 3A), in agreement with the sloppiness observations of (1).

Along each eigenvector, we then perform a *sloppy-walk* traversal: we step outward from θ^* along $\mathbf{v}_i$ in small increments and record the *neutral range* ℓ_i , defined as the greatest distance s such that $\Phi(\theta^* + s\mathbf{v}_i) = \mathbf{x}_{\text{wt}}$. A projection of a representative trajectory onto the two leading principal components of the neutral sample (Fig. 3B) shows that motion along the sloppiest direction traces a long, curved path through parameter space while the phenotype is preserved, whereas stiff-direction motion exits the neutral set within a small neighbourhood of the anchor. Quantifying this across all twelve models (Fig. 3C), the neutral range ℓ_i spans more than five orders of magnitude between the sloppiest and stiffest directions and, for the sloppiest directions, covers a substantial fraction of the full biologically plausible parameter range.

[Results paragraph: once computations are complete, give median ℓ_i for the three sloppiest eigenvectors as a fraction of the Chen-style range. Expected qualitative result: sloppy-direction traversals cover the full range; stiff-direction traversals are confined to a small neighbourhood. Comment on whether curvature of the walk indicates that the neutral set follows a curved manifold rather than a flat hyperplane.]

This result establishes that the neutral set is, at least, locally non-spherical and extended along the sloppiest Hessian directions. It does not yet tell us whether the extension continues far from the wild-type anchor, whether the manifold has non-trivial topology, or whether random points in parameter space lie close to it. For those questions we need a sampling method that does not rely on a single local expansion.

Sampling the neutral set far from the wild-type anchor. Characterising the geometry of $\mathcal{N}(\mathbf{x}_{\text{wt}})$ requires more than uniform sampling of $\mathcal{G}$. For a neutral set that occupies a small volume fraction of the full parameter space, a uniform sample populates it sparsely and non-uniformly. Thin or extended regions are under-represented, and filamentary structure, if present, is invisible. We therefore adapt the Nested Neutral Set Estimator (NNSE) of Jörg et al. (2), originally introduced to estimate neutral-set sizes for RNA molecules in continuous sequence spaces. We use it not for size estimation but for

[FIGURE PLACEHOLDER — Fig. 3: Sloppy walk results. (A) Eigenvalue spectrum of the prediction-error Hessian for the budding yeast cell-cycle model (log scale, showing ~ 8 orders of magnitude). (B) 2D PCA projection of a sloppy-direction trajectory starting at the wild-type anchor and moving outward along $\mathbf{v}_1$, compared to a stiff-direction trajectory along $\mathbf{v}_N$. Colour indicates whether the point is in the wild-type neutral set. The sloppy trajectory traces a long curved path within the neutral set; the stiff trajectory exits the set within a small neighbourhood. (C) Neutral range ℓ_i versus eigenvalue rank i for all 12 models, showing that low-eigenvalue directions admit traversals of order the full parameter range.]

Fig. 3. Sloppy directions admit long neutral traversals. (A) Eigenvalue spectrum of the prediction-error Hessian at θ^* for the budding yeast cell-cycle model (3), spanning several orders of magnitude and exhibiting the characteristic sloppy distribution. (B) Two-dimensional PCA projection of a sloppy-direction trajectory (blue) along the smallest-eigenvalue eigenvector $\mathbf{v}_1$ and a stiff-direction trajectory (orange) along the largest-eigenvalue eigenvector $\mathbf{v}_N$. The sloppy trajectory stays in the wild-type neutral set over a long curved path; the stiff trajectory leaves it within a small neighbourhood. (C) Neutral range ℓ_i along eigendirection i as a function of eigenvalue rank, for all twelve models (one curve per model). Sloppy directions (left) permit traversal over a substantial fraction of the biologically plausible range; stiff directions (right) are confined.

geometry characterisation: by seeding new sampling shells from accepted boundary points, NNSE follows the neutral set wherever it extends and returns a point cloud that reflects its shape.

Starting from θ^* , NNSE maintains a frontier $\mathcal{F}$ of accepted points at the edge of the explored region. At each iteration, a frontier point θ_k is selected and a proposal is generated as

$$\theta_{k+1} = \theta_k + r_k \mathbf{u}_k, \quad \mathbf{u}_k \sim \mathcal{D}_\lambda(\theta_k), \quad [4]$$

where r_k is a step radius drawn from an adaptive schedule and $\mathcal{D}_\lambda$ is a distribution over unit vectors concentrated in the subspace spanned by Hessian eigenvectors with eigenvalues below a threshold λ . The proposal is accepted if $\Phi(\theta_{k+1}) = \mathbf{x}_{\text{wt}}$. Accepted proposals join the sample and the frontier; rejected proposals shrink r_k locally.

[Add pseudocode / algorithm environment here.]

NNSE is *manifold-following* by construction: because new exploration is seeded from accepted points, it naturally traverses connected components and recovers extended or thread-like structures that would be missed by uniform sampling. That same property is also a concern: an algorithm that follows threads tends to find threads. The validation subsection below addresses this directly, and the topology results that follow are compared against matched-volume uniform null samples.

(Optional) Validating NNSE on controlled benchmarks. (Optional but recommended.) To test whether the topology recovered by NNSE reflects the geometry of the underlying set or is an artifact of the algorithm, we run NNSE on synthetic neutral sets of known geometry embedded in $\mathbb{R}^N$ at dimensions matching the twelve models. We use three controls: (i) a compact Gaussian blob with matched volume; (ii) an elongated ellipsoid with anisotropy ratios set by the Hessian spectrum of a representative model; and (iii) a synthetic filamentary

[FIGURE PLACEHOLDER — Fig. 4: NNSE validation on synthetic neutral sets. Three panels, one per control geometry, showing (top) the ground-truth set, (middle) the NNSE sample, and (bottom) the H_0, H_1 persistence diagrams. Blob and ellipsoid controls show short-lived H_1 features; filamentary control shows persistent H_1 . Expected conclusion: NNSE-induced topology closely tracks the underlying geometry.]

Fig. 4. (Optional.) NNSE recovers the topology of synthetic neutral sets of known geometry. Three synthetic neutral sets (compact blob, elongated ellipsoid, filamentary tubular neighbourhood) are embedded in $\mathbb{R}^N$ with N matched to a representative model. For each, NNSE samples are compared to uniform samples of the same size via persistent homology. The persistent-homology signatures of NNSE samples track the underlying geometry, arguing that the topology reported for the biological neutral sets below is not an algorithmic artifact.

set constructed as the tubular neighbourhood of a random walk in $\mathbb{R}^N$. For each, we compare the persistent-homology signatures of NNSE samples against those of uniform samples of the same size.

[Results paragraph: expected outcome is that NNSE recovers the ground-truth topology in all three cases, that H_1 persistence is small for blob and ellipsoid, and large for the filamentary control. If NNSE produces large H_1 on the blob control, that would undermine the topology claim for real data; this subsection therefore functions as a reviewer-facing check.]

The global topology of the wild-type neutral set. Traversal along sloppy eigendirections probes the neutral set one direction at a time. To characterise its *global* topology, we apply persistent homology (37, 38) to the NNSE point clouds for all twelve models. Persistent homology tracks topological features (connected components, loops, higher-dimensional voids) as a filtration parameter ϵ is varied, and reports each feature as a pair (birth, death). Features with long lifespans are interpreted as genuine topological structure; short-lived features are attributed to noise and sampling.

Signatures of localized versus extended geometry. A compact, simply-connected neutral set produces a persistence diagram dominated by a single long-lived H_0 feature and few long-lived H_1 features, and its point cloud is approximately isotropic under principal-component analysis. An extended, manifold-like neutral set produces instead (i) multiple long-lived H_0 features at small ϵ , collapsing to one as ϵ reaches the characteristic thread-joining scale; (ii) persistent H_1 features wherever extended pieces reconnect after branching; and (iii) strong anisotropy, with one or a few principal-component directions containing most of the variance.

We compare each NNSE sample against a matched-volume uniform null in the same parameter box. Across the twelve models, the wild-type NNSE samples exhibit systematically greater H_1 persistence and larger elongation than the null (Fig. 5), consistent with extended rather than localized geometry.

[Results paragraph: report summed H_1 persistence, number of persistent H_0 components at small ϵ , and PC1:PCK variance ratio, for each of the 12 models, against matched null. Expected quantitative message: H_1 persistence is $O(10\times)$ larger than null across models;

[FIGURE PLACEHOLDER — Fig. 5:
 Persistent-homology analysis of NNSE samples. (A)
 Persistence diagram (H_0 and H_1) for the budding
 yeast cell-cycle model. Off-diagonal H_1 points
 indicate persistent loops. (B) Summed H_1 persistence
 for all 12 models (coloured) versus matched-volume
 uniform null (grey). (C)
 First-three-principal-component projection of NNSE
 samples for three representative models, illustrating
 the extended, elongated structure.]

Fig. 5. Wild-type neutral sets show extended topological structure. (A) Persistence diagram (H_0 and H_1) for the wild-type neutral set of the budding yeast cell-cycle model, computed on the NNSE point cloud. Points well above the diagonal correspond to persistent topological features. (B) Total H_1 persistence for all twelve models (coloured bars) compared against a matched-volume uniform null sample (grey). NNSE samples of wild-type neutral sets consistently carry more persistent H_1 than the null. (C) PCA projection of NNSE samples for three representative models onto the first three principal components. The sampled clouds are markedly anisotropic, consistent with extension along a small number of directions.

elongation ratio > 10 . Clearly distinguish what we measure (topology of the sampled cloud) from what we infer (geometry of the manifold), with the validation subsection above as the bridge.]

Taken together with the sloppy-walk results above, this establishes that $\mathcal{N}(\mathbf{x}_{\text{wt}})$ extends far from the wild-type anchor, along many directions simultaneously, and carries topological structure that a localized compact region cannot produce.

(Optional) Reachability from random starting points. (Optional but strongly recommended.) A topology argument tells us about the shape of the neutral set; it does not directly tell us whether random points in parameter space lie close to it. Yet the latter is what the calibration paradox is about. We therefore ask a reachability question. For each of the twelve models we draw M random parameter vectors uniformly from $\mathcal{G}$ and, for each, compute the distance (in the log-parameter metric) to the nearest point in the NNSE sample of $\mathcal{N}(\mathbf{x}_{\text{wt}})$. We compare this distribution against a null in which the neutral set is a ball of the same total volume, centered at θ^* .

[Results paragraph: expected message is that the real-manifold distances are substantially smaller than the matched-volume-ball null, quantifying the accessibility advantage conferred by extended geometry. Report median distance and the fraction of random starts whose distance is within a typical Hessian-based fitting radius.]

Discussion

A layered resolution of the calibration paradox. The naive estimate in Eq. (1) implicitly assumes that parameter space is a uniform target and that the wild-type region is a single localized object to be hit by random draws. Both assumptions fail, and our results fit into a three-part picture of why. Sloppiness (1, 23) fails the first assumption locally: around any good fit, the iso-fit surface is elongated, and the acceptable region has far more volume than an isotropic estimate would suggest. Simplicity bias (27, 31) fails the first assumption globally: phenotypes are not equally likely, and simpler phenotypes, which include most biologically observed ones, occupy

[FIGURE PLACEHOLDER — Fig. 6: Reachability from random starting points. For each of the 12 models, histogram of log-parameter distance from random starts to NNSE sample (blue), compared against matched-volume-ball null (grey). Insets or summary panel: median distance and fraction within a typical fitting radius, for the real neutral set and for the null.]

Fig. 6. (Optional.) Random parameter vectors lie closer to the wild-type neutral set than to a matched-volume ball. For each model, random parameter vectors are drawn uniformly from $\mathcal{G}$ and the log-parameter distance to the nearest NNSE sample point is computed (blue histograms). The corresponding distribution for a matched-volume ball centered at the wild type is shown in grey. The systematic shift towards smaller distances in the real case is a direct measure of the accessibility advantage conferred by extended geometry.

an inflated share of parameter space. The geometry results here fail the second assumption: the wild-type neutral set is not a single localized region, but is extended along many sloppy directions and carries topological structure consistent with a space-spanning manifold. Together these three observations reduce the calibration problem from "find a specific tiny target" to "reach a connected extended structure that most starting points are already close to".

Local sloppiness and global geometry. Sloppiness and neutral-set geometry are best understood as complementary. Sloppy directions at θ^* are, to first order, tangent to the neutral manifold at that point: moving along them preserves the phenotype locally. The traversal results here extend this local picture: at least for the sloppiest directions, motion in θ -space remains in the neutral set over long distances and the manifold curves away from a pure linear subspace, which is why the sloppy-walk trajectories in Fig. 3B are curved rather than straight. Local sloppiness can be thought of as the differential structure of an extended manifold at a single point; the NNSE-plus-topology results describe what happens when that local structure is integrated out over the whole space.

Designability and arrangement. Simplicity bias and neutral-set geometry are also complementary, but for a different reason. Designability tells us that the wild-type neutral set is bigger than naive. It does not tell us how that volume is distributed. Two neutral sets of the same volume can have very different accessibility: a compact region occupying a small corner of parameter space is hard to reach, while an extended structure of the same volume is close to most starting points. Our results indicate that the wild-type neutral set is of the second type. This distinction matters especially because the revised designability picture places the wild type in the top percentile of neutral-set sizes rather than the first place; designability alone would still leave a large residual unexplained, which the geometry results address.

Evolutionary consequences. In the RNA genotype-phenotype map literature, extended neutral networks in sequence space are known to permit neutral drift, preserve function under mutation, and bring new phenotypes within reach without catastrophic loss of the existing one (28–31). Our results suggest the parameter space of GRN ODE models has a broadly analogous structure in the biochemical-parameter dimension: a population under stabilizing selection for wild-type behaviour can, in principle, drift through a large portion of parameter

space without leaving the neutral set. This offers a mechanistic substrate for canalization-like robustness (42) at the level of biochemical rate constants. We emphasize, however, that real evolution of gene regulatory networks is not confined to parametric variation; topological rearrangements of the network also occur, and a fuller treatment would need to address both axes jointly (34).

Limitations. Several caveats apply. First, our phenotype definition is the up-down oscillation string of the key output molecule; finer-grained phenotype definitions would produce smaller neutral sets and potentially a more fragmented geometry. Second, NNSE follows the manifold by construction, so the topology of its sample reflects both the underlying geometry and the algorithm's traversal. The validation subsection (if included) addresses this on synthetic controls; we nevertheless interpret the topology results as a statement about the sampled cloud first, and about the manifold second. Third, our analysis is performed at fixed network topology; biological evolution of GRNs includes the gain and loss of interactions, which changes the dimensionality of the parameter space itself. Fourth, the twelve models in our benchmark cover a range of biological contexts but are not a representative sample of all GRNs.

Outlook. The picture that emerges from sloppiness, designability and geometry together is that systems-biology parameter spaces are structured in all three senses: locally elongated, globally biased towards simple phenotypes, and arranged so that the wild-type neutral set reaches through parameter space rather than sitting in a corner of it. Each of these observations is, on its own, insufficient to explain why fitting works. In combination they make it unsurprising. The tools used here, sloppy-direction traversal, NNSE-based sampling, and persistent-homology analysis, are applicable to any parameterized input-output map, and we expect similar geometric structure to be useful in understanding other high-dimensional calibration problems in biology.

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